

EDUCATION**Nanyang Technological University, Singapore****January 2025 - present***Ph.D.* candidate, Medicine (GPA: 4.88/5.0)

Advisors: Asst. Prof. Jue Tao Lim, Dr. Ian Wee Liang En (Singapore General Hospital)

Major Collaborators: Prof. David N. Fisman (University of Toronto), Assoc. Prof. Joseph Lewnard (University of California, Berkeley), Asst. Prof. Lilith K. Whittles (Imperial College London), Asst. Prof. Sen Pei (Columbia University)

- Research interests: Systems Modeling, Statistical Inference, Deep Learning.
- [[Google Scholar](#)].

National University of Singapore, Singapore**August 2019 - June 2023***B.Eng. (Hons)*, Computer Engineering (Distinction, GPA: 4.35/5.0)

Advisors: Dr. Ganesh Neelakanta Iyer, Dr. Ravi Suppiah, Asst. Prof. Deepu John (University College Dublin)

- University Town College Program

PUBLICATIONS

1. Lin Geng*, Perran A. Ross*, Jo Yi Chow, **Zihao Wang**, Esther Li Wen Choo, Chia-Chen Chang, Xinyue Gu, Ary Hoffmann, Jue Tao Lim, “*Wolbachia*-mediated incompatible-insect technique robust to future climate change scenarios,” submitted.
 2. An Ting Tay*, Peihong Guo*, Jinghao Nicholas Ngaim*, Zihao Wang*, Liang En Wee, Calvin Chiew, Lalitha Kurupatham, Lee Ching Ng, Po Ying Chia, David Chien Boon Lye, Kelvin Bryan Tan, Jue Tao Lim*, “Risk of post-acute multi-organ sequelae after dengue reinfection,” submitted.
 3. Liang En Wee*, Reen Wan Li Ho*, **Zihao Wang***, Peihong Guo*, Rhea Khanna, Calvin Chiew, Russell Jingxian Li, Jiahui Li, Chee-Fu Yung, Chia Yin Chong, Zubaidah Said, David Chien Boon Lye, Vernon Lee, Kelvin Bryan Tan, Jue Tao Lim, “Acute and post-acute healthcare utilization following paediatric hospitalization for respiratory-syncytial-virus versus influenza or Omicron COVID-19 in a tropical setting: a population-based cohort study,” submitted.
 4. Peihong Guo*, Reen Wan Li Ho*, **Zihao Wang***, Ian Liang En Wee*, Russ Li, David Chien Boon Lye, Kelvin Bryan Tan, Jue Tao Lim, “Acute and post-acute healthcare utilization after hospitalization for COVID-19, influenza or respiratory syncytial virus in adults: a population-based cohort study,” submitted.
 5. **Zihao Wang**, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, “Long-term public health impact of doxycycline post-exposure prophylaxis on syphilis transmission,” *Nature Health*, 1-13, 2026. [[Paper](#), [GitHub](#)].
 6. **Zihao Wang**, Yuhao Li, Yao Shi, and Zhe Wu, “Input convex lipschitz recurrent neural networks for robust and efficient process modeling and optimization,” arXiv. [[arXiv](#), [GitHub](#)].
 7. **Zihao Wang**, and Zhe Wu, “Towards foundation model for chemical reactor modeling: Meta-learning with physics-informed adaptation,” *Chemical Engineering Research and Design* 218, 839-853, 2025. IF: 3.9, JCR Q2 (Engineering, Chemical). [[Paper](#), [GitHub](#)].
 8. **Zihao Wang**, Donghan Yu, and Zhe Wu, “Real-time machine-learning-based optimization using input convex long short-term memory network,” *Applied Energy* 377, 124472, 2025. IF: 11.0, JCR Q1 (Engineering, Chemical; Energy & Fuels). [[Paper](#), [GitHub](#)].
 9. **Zihao Wang**, and Ravi Suppiah, “Upper limb movement recognition utilising EEG and EMG signals for rehabilitative robotics,” *Proceedings of Future of Information and Communication Conference*, 676-695, San Francisco, California, USA, 2023. [[Paper](#), [GitHub](#)].
- (* denotes equal contribution)

PRESENTATIONS

1. **Zihao Wang***, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, “Long-term public health impact

of doxycycline post-exposure prophylaxis on syphilis transmission,” *Singapore HIV, Hepatitis and Sexually Transmitted Infections Congress (SHHSC)*, Singapore, 2026. **(Oral)**.

2. **Zihao Wang***, Dariya Nikitin, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, “Modelling the interplay between doxycycline post-exposure prophylaxis, vaccination, and antimicrobial resistance in *Neisseria gonorrhoeae*,” *European Congress of Clinical Microbiology and Infectious Diseases (ESCMID Global 2026)*, Munich, Germany, 2026. (Poster, **Top 2%**).
3. **Zihao Wang***, Dariya Nikitin, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, “Modelling the interplay between doxycycline post-exposure prophylaxis, vaccination, and antimicrobial resistance in *Neisseria gonorrhoeae*,” *LKCMedicine Early Researcher Network (LEARN) Symposium*, Singapore, 2026. **(Flash Talk)**.
4. **Zihao Wang***, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, “Long-run public health impact of doxycycline post-exposure prophylaxis and behavioural factors on syphilis transmission: A modelling study in Singapore and England,” *Epidemics: 10th International Conference on Infectious Disease Dynamics (Epidemics10)*, San Diego, California, USA, 2025. **(Oral)**.
5. **Zihao Wang***, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, “Long-term public health impact of doxycycline post-exposure prophylaxis and behavioural factors on syphilis transmission: A modelling study in Singapore and England,” *Singapore Health & Biomedical Congress (SHBC)*, Singapore, 2025. (Poster).
6. **Zihao Wang**, Wenlong Wang*, and Zhe Wu, “Accelerated modeling of various chemical processes using meta-learning-based foundation models: A few-shot learning approach using Reptile,” *American Institute of Chemical Engineers (AIChE) Annual Meeting*, San Diego, California, USA, 2024. **(Oral)**. [[Abstract](#)].
7. **Zihao Wang**, Wenlong Wang*, and Zhe Wu, “Input convex LSTM for fast machine learning-based optimization,” *American Institute of Chemical Engineers (AIChE) Annual Meeting*, San Diego, California, USA, 2024. (Poster). [[Abstract](#)].
8. **Zihao Wang**, and Zhe Wu*, “Input convex LSTM: A convex approach for fast model predictive control,” Institute for Machine Learning (Head: Professor Sepp Hochreiter), Johannes Kepler University, Linz, Austria, December 2023. **(Invited talk by Professor Günter Klambauer)**.
(* denotes presenting author)

AWARDS AND HONORS

Travel Grant, ESCMID Global 2026	04/2026
Top Rated Abstract, ESCMID Global 2026	04/2026
NTU Research Scholarship, NTU	2025 - Present
IEEE Singapore Computer Society Prize, NUS	06/2023
Outstanding Computing Project Prize, NUS	06/2023

DISSERTATIONS

1. **Zihao Wang**, “Multimodal sensor fusion using AI for healthcare IoT sensors,” B.Eng. Dissertation, Computer Engineering, National University of Singapore, Singapore, 2023. [[Paper](#)].
2. **Zihao Wang**, “AI research for rehabilitative robotics,” Undergraduate Research Opportunity Project (UROP) Thesis, Computer Engineering, National University of Singapore, Singapore, 2022. [[Paper](#)].

PROFESSIONAL EXPERIENCES

National Centre for Infectious Diseases	May 2025 - May 2026
<i>Research Staff (Part-time)</i> , Tan Tock Seng Hospital, Singapore	
Advisors: Assoc. Prof. Kelvin Bryan Tan, Asst. Prof. Jue Tao Lim	
Conduct research on evaluating different methodological approaches to estimate the true infection fatality ratio (IFR) during pandemics, with the aim of informing early estimation of IFR under limited data to support timely,	

evidence-based policy decisions, under the Programme for Research in Epidemic Preparedness and REsponse (PREPARE)

- Conduct research on population-based retrospective cohort studies for respiratory syncytial virus (RSV) and dengue

Department of Chemical & Biomolecular Engineering

July 2023 - January 2025

Research Engineer (Full-time), National University of Singapore, Singapore

Advisor: Asst. Prof. Zhe Wu

- Conducted research on input convex neural networks, model predictive control, chemical process modeling, meta-learning, and physics-informed learning
- Wrote concept papers and full proposals for grant applications and supervised master's students

Department of Biomedical Engineering

March 2023 - July 2023

Undergraduate Research Assistant (Internship), National University of Singapore, Singapore

Advisor: Asst. Prof. Yueming Jin

- Conducted research on computer vision for gastrectomy surgical video analysis and modality-cataract video segmentation

Rolls-Royce@NTU Corporate Laboratory

May 2022 - August 2022

Research Engineer (Internship), Nanyang Technological University, Singapore

Advisors: Dr. Aditya Venkataraman and Dr. Benjamin Cheong

- Worked on software engineering solutions for hybrid-electric power and propulsion design tools for marine and aero engines

NUS Human-Computer Interaction (NUS-HCI) Lab

March 2022 - May 2022

Undergraduate Research Assistant (Internship), National University of Singapore, Singapore

Advisor: Prof. Shengdong Zhao

- Conducted empirical research on human-computer interaction

SKILLS

Computer: C, C++, Java, Python, R, Assembly Language, MATLAB, PostgreSQL, Verilog, Vensim, Gephi, LaTeX

Languages: Mandarin (native), English (bilingual), Cantonese (proficient), Korean (conversational)